

GitHub → Jenkins → Docker CI/CD Pipeline (Step-by-Step)

Step 1: Prepare GitHub Repository

Ensure your repository contains a Dockerfile and index.html file.

```
FROM nginx:latest
RUN rm -rf /usr/share/nginx/html/*
COPY index.html /usr/share/nginx/html/index.html
```

Step 2: Setup Jenkins

Install Jenkins and required plugins: Git, GitHub Integration.

Step 3: Give Docker Permission to Jenkins

```
sudo usermod -aG docker jenkins
sudo systemctl restart jenkins
```

Step 4: Create Freestyle Job

Create a new Freestyle project in Jenkins named 'web-server'.

Step 5: Connect GitHub Repository

```
https://github.com/<your-username>/<your-repo>.git
```

Step 6: Enable Webhook Trigger

Enable 'GitHub hook trigger for GITScm polling' in Build Triggers.

Step 7: Add Docker Build Commands

```
docker stop myapp || true
docker rm myapp || true
docker build -t myapp .
docker run -d -p 8080:80 --name myapp myapp
```

Step 8: Configure GitHub Webhook

Payload URL:
`http://<your-server-ip>:8080/github-webhook/`

Content type:
`application/json`

Event:

Just the push event

Step 9: Open Required Ports

Allow ports 8080 (Jenkins) and 8000 (application) in your server firewall.

Step 10: Test Pipeline

```
git add .  
git commit -m "test webhook"  
git push origin main
```

Access Application

`http://<your-server-ip>:8000`